



Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404

Status Update 5

Group #17: Patent Miner

List of team members:

Christian Casteel

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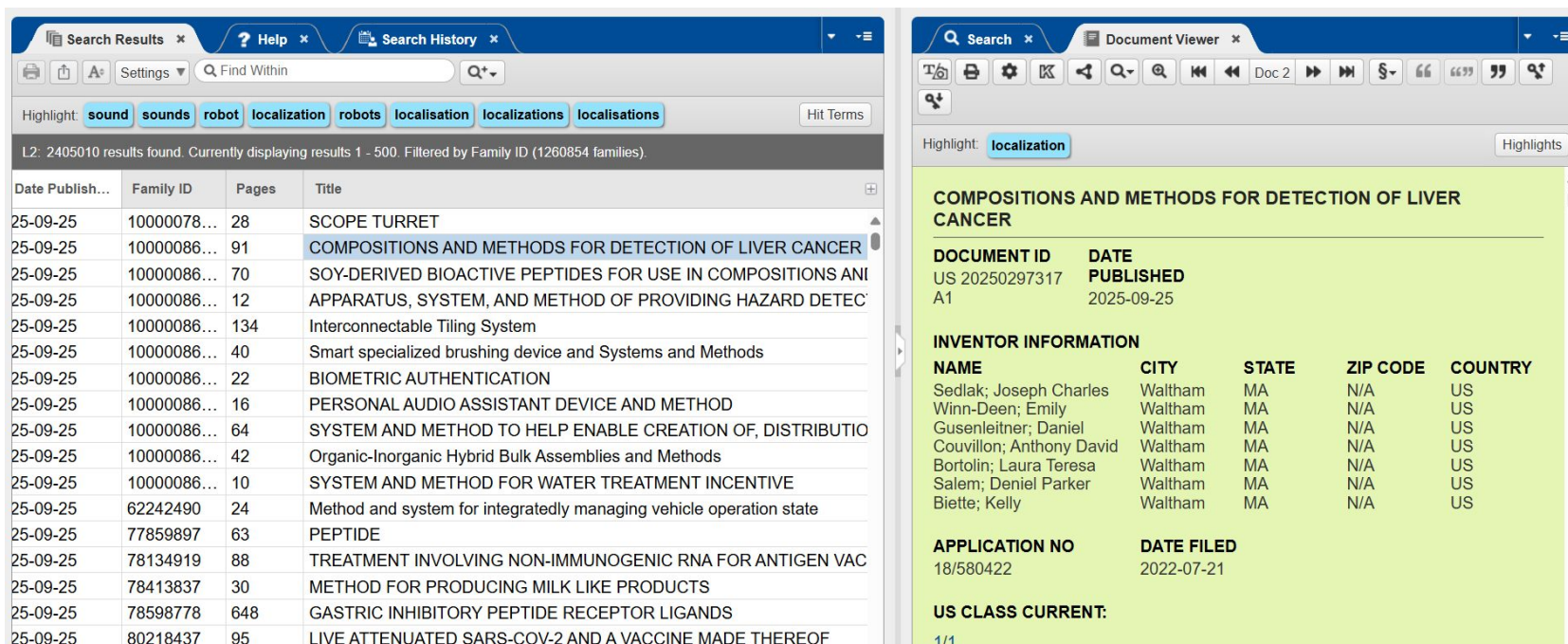
Project Summary

Problem statement:

- The current keyword search implemented by the United States Patent and Trademark Office (USPTO) fails to capture contextual meaning, leading to inaccuracy of search results and expensive billing rates for litigation professionals.

Solution:

- Development of a patent database mining tool that optimizes the patent search process by retrieving contextually relevant patents faster and more accurately compared to a traditional database.



The screenshot displays a patent search interface. The left pane shows search results for the keyword 'localization'. The right pane shows the detailed view of the selected patent, 'COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER'.

Search Results (Left Pane):

Date Publish...	Family ID	Pages	Title
25-09-25	10000078...	28	SCOPE TURRET
25-09-25	10000086...	91	COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER
25-09-25	10000086...	70	SOY-DERIVED BIOACTIVE PEPTIDES FOR USE IN COMPOSITIONS ANI
25-09-25	10000086...	12	APPARATUS, SYSTEM, AND METHOD OF PROVIDING HAZARD DETEC
25-09-25	10000086...	134	Interconnectable Tiling System
25-09-25	10000086...	40	Smart specialized brushing device and Systems and Methods
25-09-25	10000086...	22	BIOMETRIC AUTHENTICATION
25-09-25	10000086...	16	PERSONAL AUDIO ASSISTANT DEVICE AND METHOD
25-09-25	10000086...	64	SYSTEM AND METHOD TO HELP ENABLE CREATION OF, DISTRIBUTIO
25-09-25	10000086...	42	Organic-Inorganic Hybrid Bulk Assemblies and Methods
25-09-25	10000086...	10	SYSTEM AND METHOD FOR WATER TREATMENT INCENTIVE
25-09-25	62242490	24	Method and system for integratedly managing vehicle operation state
25-09-25	77859897	63	PEPTIDE
25-09-25	78134919	88	TREATMENT INVOLVING NON-IMMUNOGENIC RNA FOR ANTIGEN VAC
25-09-25	78413837	30	METHOD FOR PRODUCING MILK LIKE PRODUCTS
25-09-25	78598778	648	GASTRIC INHIBITORY PEPTIDE RECEPTOR LIGANDS
25-09-25	80218437	95	LIVE ATTENUATED SARS-COV-2 AND A VACCINE MADE THEREOF

Document View (Right Pane):

COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER

DOCUMENT ID	DATE
US 20250297317 A1	PUBLISHED 2025-09-25

INVENTOR INFORMATION

NAME	CITY	STATE	ZIP CODE	COUNTRY
Sedlak; Joseph Charles	Waltham	MA	N/A	US
Winn-Deen; Emily	Waltham	MA	N/A	US
Gusenleithner; Daniel	Waltham	MA	N/A	US
Couvillon; Anthony David	Waltham	MA	N/A	US
Bortolin; Laura Teresa	Waltham	MA	N/A	US
Salem; Deniel Parker	Waltham	MA	N/A	US
Biette; Kelly	Waltham	MA	N/A	US

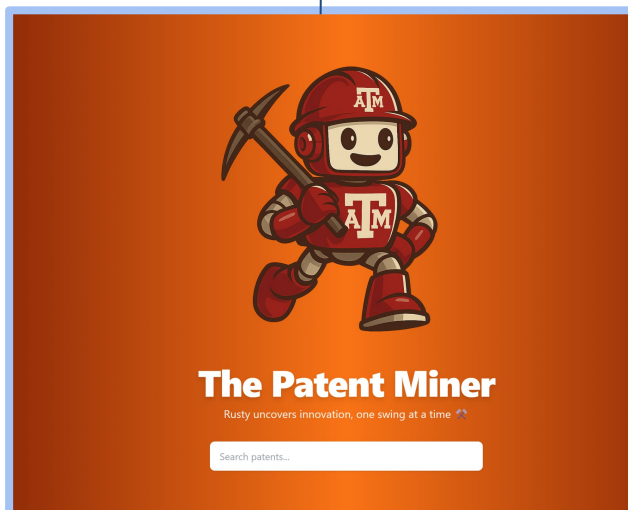
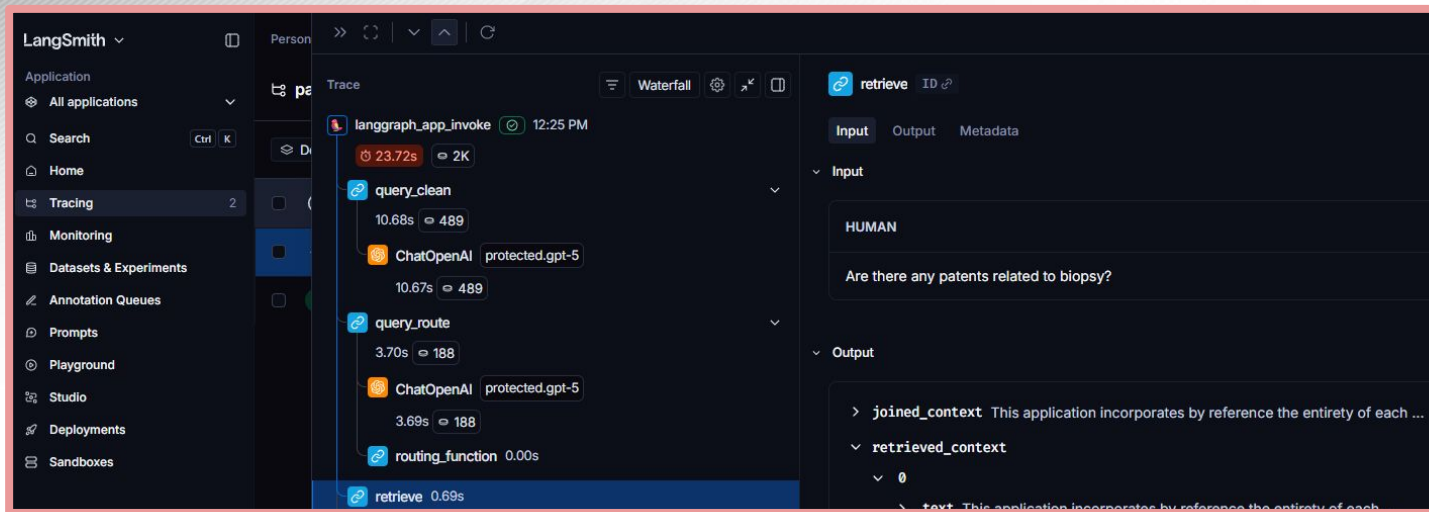
APPLICATION NO 18/580422 **DATE FILED** 2022-07-21

US CLASS CURRENT: 1/1

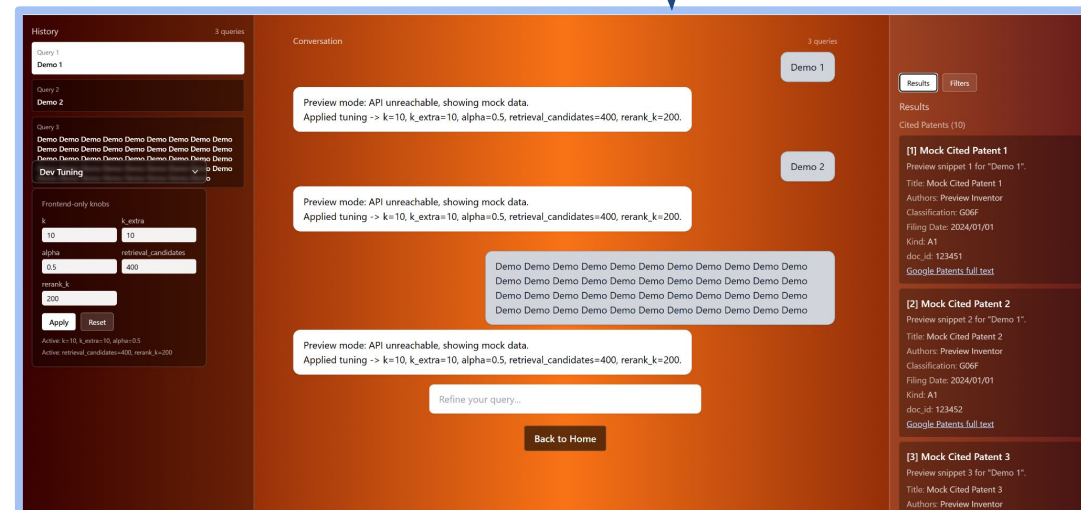
System Overview

Kosi Ezewudo

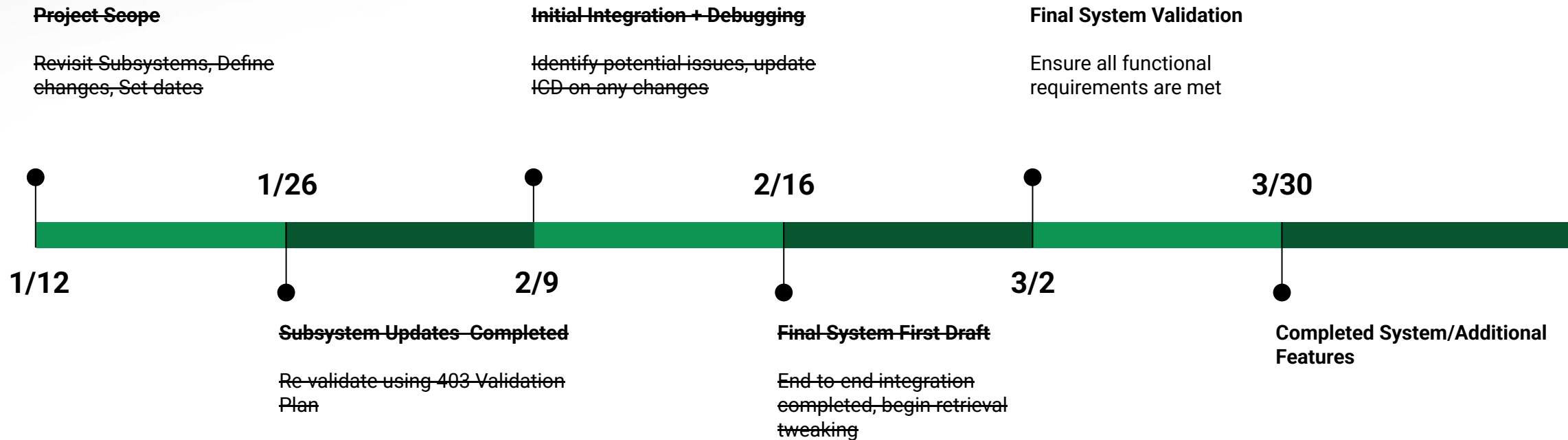
Christian Casteel



CJ Vines



Project Timeline



Embedding Pipeline and Storage

Christian Casteel

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation
• Reformatted results and conversation history to frontend • Added dev tuning and laid groundwork for metadata filtering to frontend	• Create fake patents for failure compliance test and run validation • Verify metadata integrity • Complete full patent validation • Find prior art patents, validate prior art search capabilities*

Embedding Pipeline and Storage

Christian Casteel

The image displays a web application for a patent search system, organized into three main vertical sections: a sidebar on the left, a central conversation area, and a results panel on the right.

The sidebar on the left is dark-themed and contains:
- A 'History' section at the top with a '3 queries' indicator, showing three queries: 'Query 1 Demo 1', 'Query 2 Demo 2', and 'Query 3' (which is expanded to show a list of 'Demo' items and a 'Dev Tuning' option with a dropdown arrow).
- A 'Frontend-only knobs' section with four input fields: 'k' (set to 10), 'k_extra' (set to 10), 'alpha' (set to 0.5), and 'retrieval_candidates' (set to 400). Below these is a 'rerank_k' field (set to 200) and 'Apply' and 'Reset' buttons.
- A status bar at the bottom indicating 'Active: k=10, k_extra=10, alpha=0.5' and 'Active: retrieval_candidates=400, rerank_k=200'.

The central conversation area has a light gray background and is titled 'Conversation' with a '3 queries' indicator. It features a chat interface with:
- Three white message bubbles, each containing the text: 'Preview mode: API unreachable, showing mock data. Applied tuning -> k=10, k_extra=10, alpha=0.5, retrieval_candidates=400, rerank_k=200.'
- A light blue input box at the bottom with the placeholder text 'Refine your query...'.
- A 'Back to Home' button at the very bottom center.

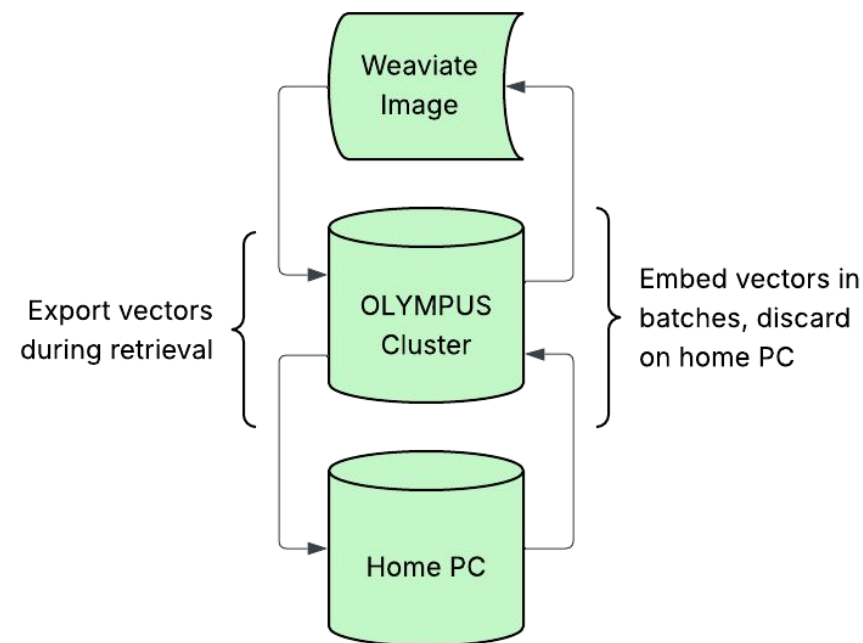
The results panel on the right is light gray and contains:
- A 'Results' and 'Filters' toggle at the top.
- A 'Filter current results (frontend only)' section with input fields for 'Author' (e.g. smith), 'CPC / Classification' (e.g. G06F), and 'doc_id contains' (e.g. 12345).
- 'Filing from' and 'Filing to' date pickers (mm/dd/yyyy).
- 'Kind' and 'Claim type' dropdown menus (both set to 'Any').
- A 'Clear Filters' button.
- A list of results, each starting with a title (e.g., '[1] Mock Cited Patent 1'), followed by a 'Preview snippet' and a list of metadata: 'Title: Mock Cited Patent 1', 'Authors: Preview Inventor', 'Classification: G06F', 'Filing Date: 2024/01/01', 'Kind: A1', and 'doc_id: 123451'. Each result also includes a link to 'Google Patents full text'.
- The visible results are '[1] Mock Cited Patent 1', '[2] Mock Cited Patent 2', and '[3] Mock Cited Patent 3', each with its own preview snippet and metadata.

Frontend/Preprocessing CJ Vines

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation
• Embedded ~60k patents • OLYMPUS Cluster space reserved • SSH and gRPC tunnel successfully implemented and documented	• Run Qrels • Scale up as high as possible on OLYMPUS Cluster • Test different MUVIRA retrieval knobs, possibly also remove PQ • Implement Metadata Filtering

OLYMPUS Cluster update

- OLYMPUS Cluster has:
 - 1 TB Shared memory (Probably need 25-50 GB max)
 - 1 TB hard drive space on scratch
 - More than enough for scaling up to ~200-250k patents
- No more OOM issues
 - More hard drive space could be desired
 - Possible room for space optimization if we rehydrate metadata from home PC, but that requires abandoning implementation of metadata filtering in a classical sense
 - Now we can optimize for accuracy rather than being memory constrained
 - Query latency is of lesser importance than accuracy for our use case



Retrieval Augmented Generation (RAG) Subsystem

Kosi Ezewudo

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation
• Full system refinement: ○ Finalized short-term memory implementation • As per-TA suggestion: ○ Implemented Context Uniqueness Layer ○ Performing validation on patent-claims-chunk level context	• Continue Full system and Sub-system refinement ○ Reference retrieved context in response • Brainstorm potential pathways for; ○ Implementation of claims defense/combat dedicated layer. • Perform extensive validation

Retrieval Augmented Generation Subsystem

Kosi Ezewudo

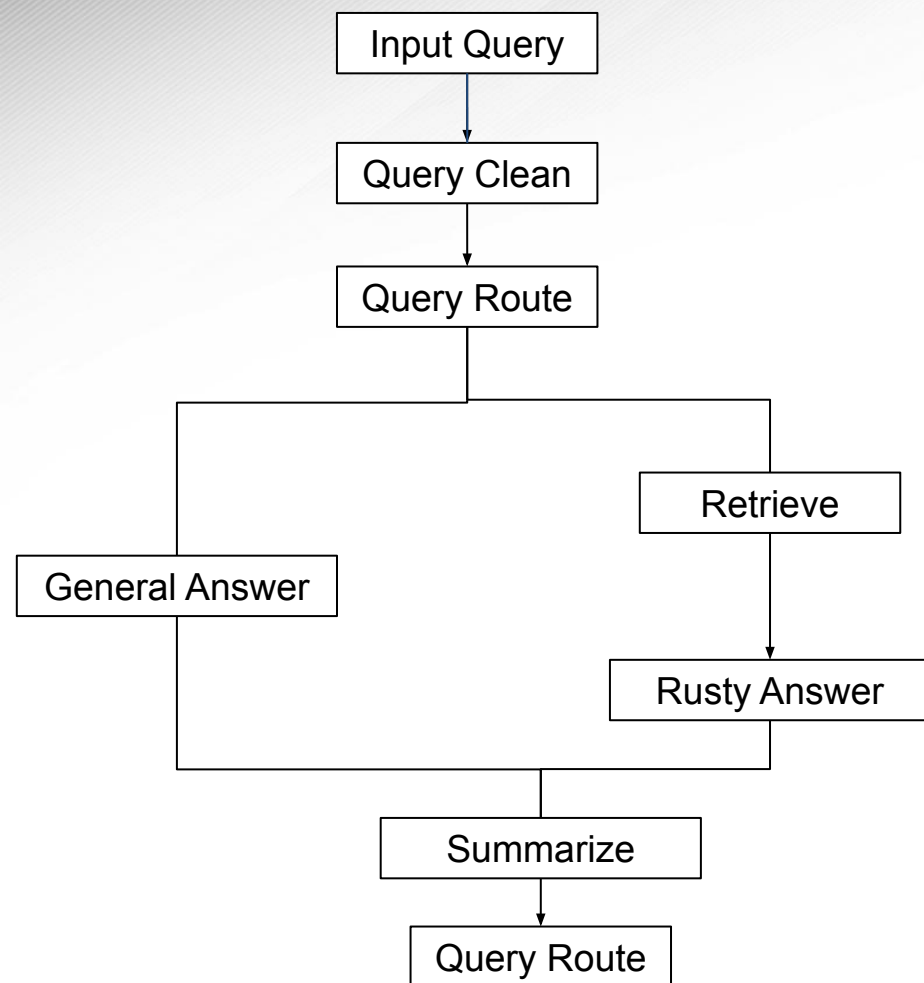


Fig 1. Updated Patent Miner RAG pipeline

Claims

1. A method of monitoring a device comprising a soft pneumatic actuator and a gas pump, the soft pneumatic actuator configured to actuate, assist and/or resist a movement of a body part, wherein the soft pneumatic actuator is in fluid communication with the gas pump, and wherein the gas pump is for pressurising and/or depressurising an internal chamber of the soft pneumatic actuator, the device further comprising a pressure sensing means for measuring the pressure of a gas in the internal chamber of the soft pneumatic actuator, the method comprising: measuring the pressure of the gas in the internal chamber of the soft pneumatic actuator, and using the pressure measurement or measurements to estimate any one or any combination of: (a) the volume of the internal chamber of the soft pneumatic actuator; (b) the three-dimensional position, shape, and/or displacement of the soft pneumatic actuator; and (c) the force delivered by the soft pneumatic actuator.
2. The method as claimed in claim 1, wherein the soft pneumatic actuator is positioned or fixed in relation to the corresponding body part of a user, and wherein the pressure measurement or measurements is/are used for any one or any combination of: (a) tracking the three-dimensional position or displacement of the body part; (b) determining the starting or resting position of the body part; (c) determining when the gas pump should be turned on and/or off or determining when pressurisation and/or depressurisation of the internal chamber should start or stop; and (d) monitoring the strength of the body part or the resistance of the body part to actuation by the soft pneumatic actuator.
3. The method as claimed in claim 2, wherein the body part is a finger and the soft pneumatic actuator is a bending actuator, and wherein the pressure measurement or measurements is/are used to track the degree of extension and/or flexion of the finger.

Fig 2. Soft Robotic Assistive Device Claims Section

Execution Plan

[illegible]

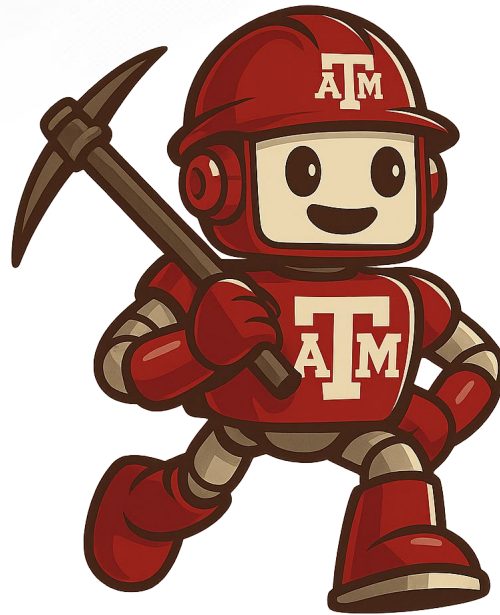
Validation Plan

Paragraph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.2.	API/Pydantic Test	API gives correct response depending on the route taken 100% of the time	Send input query and data of different variations including edge cases like empty queries, queries with special characters, and get requests of the wrong shape	Tested	CJ Vines
3.2.1.1.	Determinism Test	Preprocessing Subsystem resulting chunks consistent across runs on the same dataset	Calculate SHA-256 hash of entire list of dictionaries after chunking and compare hash across runs to ensure download pipeline is deterministic. If any run is inconsistent, the test fails	Tested	CJ Vines
3.2.1.1.	Metadata Extraction	Stored metadata is verified to be at least 99% accurate	Compare metadata after chunking to metadata called from USPTO API, aiming for 99% accuracy	Tested	CJ Vines
3.2.1.4.	Full System Stress Test	Multiple simultaneous API calls are made successfully no data loss within 1ms	Make multiple API calls simultaneously, handle each concurrently in order.	Untested	CJ Vines & Kosi Ezewudo & Christian Casteel
3.2.1.5.	Embedding Accuracy Test	"Golden" query - patent pairs must have $\geq .8$ Hit@k, Mean Reciprocal Rank (MRR) $\geq .4$, and Normalized Discounted Cumulative Gain (NDCG) of .4	k = 10 target patents at minimum with an equal number of matching queries. Calculate Hit@k, MRR, and NDCG for dataset. "Golden" set will include 50 hand selected patents	Untested	Christian Casteel
3.2.1.1.	Failure Compliance Test	Pipeline can take in incorrect patent data and continue embedding without failure. Also will flag patents with false data	5 fake patents containing empty and improperly formatted metadata will be sent through the pipeline, and will throw error flags. System should continue embedding without failure	Untested	Christian Casteel
3.2.1.6.	Metadata Integrity Test	All stored metadata must be 100% consistent with metadata listed on a patent	Select 50 patents, specifically patents with generally unique CPCs and numbers of authors, then compare all metadata fields to the same patents on the USPTO database	Untested	Christian Casteel
3.2.1.1.	Download and Embedding Pipeline Stress Test	A USPTO bulk dataset is downloaded, embedded, and stored without failure	A USPTO bulk dataset download is conducted via fronetend download script. The bulk dataset should contain minimum 1000 patents	Tested	Christian Casteel & CJ Vines
3.2.1.3.	Query Cleaning Test	Removes typos, sentence fragments etc. from user query	Execute patent search using grammatically incorrect queries	Untested	Kosi Ezewudo
3.2.1.4.	Generation Quality and Latency Test	Generated responses must have $\geq .8$ Answer relevancy score, $\geq .8$ Faithfulness score and benchmark Latency results	Query Patent Miner using golden set of patents. Generate responses and calculate Answer relevancy, Faithfulness and Response Latency	Untested	Kosi Ezewudo & Christian Casteel & CJ Vines



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Thank you!